# RangeVol: volatility from the day's high and low

The open, high, low and close of each day say more about volatility than the closes alone: six classical estimators that use them, and the log range of Alizadeh, Brandt and Diebold.

| | |
|---|---|
| **In Pollis** | The *Range-Based Volatility Estimation* panel, set aside for a future Finance Toolbox; its menu entry is commented out until then. |
| **Missing in Julia** | the range-based estimators of Parkinson, Garman and Klass, Rogers and Satchell, and Yang and Zhang, with close-to-close as the baseline, for daily OHLC prices. R has them in TTR's `volatility()`; Julia had none. |
| **Package** | `RangeVol.jl`, from GitHub: `Pkg.add(url = "https://github.com/Trumpingtons/RangeVol.jl")`. |
| **Tests** | 37 tests: 18 against TTR on three datasets, 14 on simulated prices, 5 of argument checks. Recomputed today, the six estimators agree with TTR 0.24.4 to $5.3 \times 10^{-14}$ or closer. |

## Why a package

The usual estimate of volatility is the standard deviation of daily returns, close to close. It ignores what happened during the day: a day that ends where it started may have travelled far in between. Parkinson (1980) showed that the high and low of the day, in a random walk, estimate the variance about five times as efficiently as the close; Garman and Klass (1980) added the open and the close; Rogers and Satchell (1991) allowed for a drift; Yang and Zhang (2000) allowed for the move overnight, between one close and the next open. Daily OHLC prices are what most data sources give, so these estimators come at no cost in data. Pollis' panel needed them in Julia, with the same conventions as TTR, so that a result can be checked in R.

### Where they are used

- **Historical volatility** of stocks, indices, currencies and commodities, for risk measures and option pricing, when intraday data are not available.
- **Short windows.** With five times the efficiency, a month of ranges estimates the variance about as well as five months of closes.
- **Volatility models.** The log range is a nearly Gaussian measure of volatility for fitting stochastic volatility models (Alizadeh, Brandt and Diebold 2002).

## The method

For $n$ days with open $O_t$, high $H_t$, low $L_t$ and close $C_t$, each function returns $\sqrt{\hat{\sigma}^2 \cdot k}$, where $\hat{\sigma}^2$ is the daily variance and $k$ is `trading_periods`: 1 by default, for daily volatility; 252 to annualise a stock's, as TTR's `N = 252` does.

| Function | Daily variance |
|---|---|
| `close_to_close(c)` | $\frac{1}{n-2}\sum_{t=2}^{n}(r_t - \bar{r})^2$, with $r_t = \ln(C_t/C_{t-1})$ |
| `parkinson(h, l)` | $\frac{1}{4n\ln 2}\sum_t \ln(H_t/L_t)^2$ |
| `garman_klass(o, h, l, c)` | $\frac{1}{n}\sum_t\left[\frac{1}{2}\ln(H_t/L_t)^2 - (2\ln 2 - 1)\ln(C_t/O_t)^2\right]$ |
| `rogers_satchell(o, h, l, c)` | $\frac{1}{n}\sum_t\left[\ln(H_t/C_t)\ln(H_t/O_t) + \ln(L_t/C_t)\ln(L_t/O_t)\right]$ |
| `garman_klass_yang_zhang(o, h, l, c)` | $\frac{1}{n-1}\sum_{t=2}^{n}\left[\ln(O_t/C_{t-1})^2 + \frac{1}{2}\ln(H_t/L_t)^2 - (2\ln 2 - 1)\ln(C_t/O_t)^2\right]$ |
| `yang_zhang(o, h, l, c)` | $\hat{\sigma}_o^2 + \kappa\,\hat{\sigma}_c^2 + (1-\kappa)\,\hat{\sigma}_{RS}^2$ over $t = 2, \ldots, n$, with $\kappa = 0.34/(1.34 + \frac{m+1}{m-1})$, $m = n - 1$; needs $n \geq 3$ |

In Yang and Zhang's estimator, $\hat{\sigma}_o^2$ is the sample variance of the overnight returns $\ln(O_t/C_{t-1})$, $\hat{\sigma}_c^2$ that of the open-to-close returns $\ln(C_t/O_t)$, and $\hat{\sigma}_{RS}^2$ the Rogers–Satchell variance; $\kappa$ is the weight that minimizes the variance of the estimate. Parkinson's and Garman and Klass's estimators assume no drift and no move overnight; Rogers and Satchell's allows for a drift; the last two include the overnight move, and Yang and Zhang's allows for both.

### The log range

`alizadeh_brandt_diebold(h, l)` returns a vector: the range of the log price of each day, $\ln H_t - \ln L_t$. Its logarithm, $\ln(\ln H_t - \ln L_t)$, is the log range of Alizadeh, Brandt and Diebold (2002), which is close to Gaussian with a mean that moves with $\ln \sigma_t$: take `log.(alizadeh_brandt_diebold(h, l))`. Over 100,000 simulated days of 390 steps, the range had

skewness 0.962 and kurtosis 4.176; its logarithm 0.139 and 2.758, with a mean $0.384 + \ln\sigma$ and a standard deviation of 0.299.

## A worked example

### AAPL in 2023

The 250 trading days of 2023, from Yahoo Finance through R's quantmod, are in `test/data/aapl.csv`.

```
using RangeVol, CSV
d = CSV.File("test/data/aapl.csv")
o, h, l, c = d.open, d.high, d.low, d.close
yang_zhang(o, h, l, c)                                  # 0.013443, daily
yang_zhang(o, h, l, c; trading_periods = 252)         # 0.213400, annualised
```

| **Annualised,** $k = 252$ | **close** | **Parkinson** | **G–K** | **R–S** | **GK–YZ** | **Y–Z** |
|---|---|---|---|---|---|---|
| AAPL 2023, 250 days | 0.199154 | 0.178093 | 0.178875 | 0.179683 | 0.213309 | 0.213400 |
| EUR/USD 2023, 260 days | 0.075319 | 0.065705 | 0.077361 | 0.092476 | 0.107925 | 0.113978 |

For AAPL, the three estimators that see only the day itself give about 0.179; those that add the move overnight, 0.213, above close-to-close's 0.199. The moves between one close and the next open, as earnings and news released outside trading hours would make them, add to the variance, and the first three miss them.

### What the data must be

The EUR/USD row is a warning. In Yahoo's series the close equals the open on 259 of the 260 days, so $\ln(C_t/O_t) = 0$ and the whole day's move between closes appears as a move overnight: Garman and Klass's, Rogers and Satchell's and Yang and Zhang's estimators then measure something else, and they disagree with each other and with close-to-close. The estimators assume that the open, high, low and close come from one session; check that they do before trusting a range-based estimate.

## What the tests show

### Against R

`test/generate_data.R` downloads AAPL and EUR/USD, simulates a third series, and saves TTR 0.24.4's `volatility()` of each with `N = 252`, over the whole series (`n = nrow` for the first four, `n = nrow - 1` for the two with an overnight term, whose window counts pairs of days). The estimates RangeVol computes today differ from TTR's by at most $5.3 \times 10^{-14}$, relative, in each of the 18; running TTR again in R gives the saved values to about 15 digits. The test suite compares two saved CSVs, TTR's and one RangeVol wrote, at a relative tolerance of $10^{-4}$, and does not recompute.

### By simulation

`vignette/simulation.jl` draws prices whose log moves as a random walk, with a daily standard deviation of 0.0126, in 390 steps a day (one a minute) or 23,400 (one a second), with or without a move overnight; 2,000 samples of 250 days, or 400 at seconds. The table gives the mean of each daily variance over the true one, and the efficiency: the variance of close-to-close's estimate over the estimator's, each divided by its mean.

| | **close** | **Parkinson** | **G–K** | **R–S** | **GK–YZ** | **Y–Z** |
|---|---|---|---|---|---|---|
| *Minutes, no move overnight* | | | | | | |
| mean / true | 1.003 | 0.936 | 0.910 | 0.910 | 0.910 | 0.923 |
| efficiency | 1 | 4.65 | 7.21 | 5.64 | 7.18 | 6.78 |
| *Minutes, a fifth of the variance overnight* | | | | | | |
| mean / true | 1.001 | 0.749 | 0.728 | 0.728 | 0.928 | 0.938 |
| efficiency | 1 | 4.50 | 6.96 | 5.45 | 7.61 | 7.24 |
| *Seconds, no move overnight* | | | | | | |
| mean / true | 1.001 | 0.994 | 0.991 | 0.991 | 0.991 | 0.992 |
| efficiency | 1 | 4.68 | 6.98 | 5.98 | 6.95 | 6.81 |

The efficiencies are close to those the papers derive for continuous prices, about 5.2 for Parkinson's, 7.4 for Garman and Klass's and 6 for Rogers and Satchell's. A high and low observed once a minute fall short of the true ones, and the range-based estimators then understate the variance by 6 to 9 per cent; once a second, by less than 1. With a move overnight, the first three miss it, as in AAPL, and only the last two stay close to the truth.

### The rest

`test/runtests.jl`: 14 tests that each estimator is positive and near the volatility of a simulated series, and 5 that wrong arguments (vectors of different lengths, too few days) throw. The simulated highs and lows in these tests are

drawn around the open and close, not from a path, and the tolerance of 0.005 is wide against a daily volatility of 0.0126; `vignette/simulation.jl` is the stronger check.

## Still open

- **Recompute in the tests.** The comparison with TTR should compute RangeVol's estimates in the test, from the data, rather than read them from a CSV written earlier.
- **Rolling windows.** TTR's `volatility()` returns an estimate for each window of $n$ days; RangeVol returns one for the whole series.
- **Bias corrections.** The discrete-sampling bias above has known corrections, and Yang and Zhang's $\kappa$ assumes that the drift is small; neither is offered.
- **The name of the log range.** `alizadeh_brandt_diebold` returns the range, whose logarithm is Alizadeh, Brandt and Diebold's log range; the function's name and docstring should say so.

## References


Alizadeh, S., Brandt, M. W. and Diebold, F. X. (2002). Range-based estimation of stochastic volatility models. *Journal of Finance* 57(3), 1047–1091. doi:10.1111/1540-6261.00454

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Software: Ulrich, J., TTR and quantmod (R); CSV.jl, Tables.jl, StableRNGs.jl and Statistics (MIT).



Package written and tested with Claude Code (Anthropic) in Pollis.


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